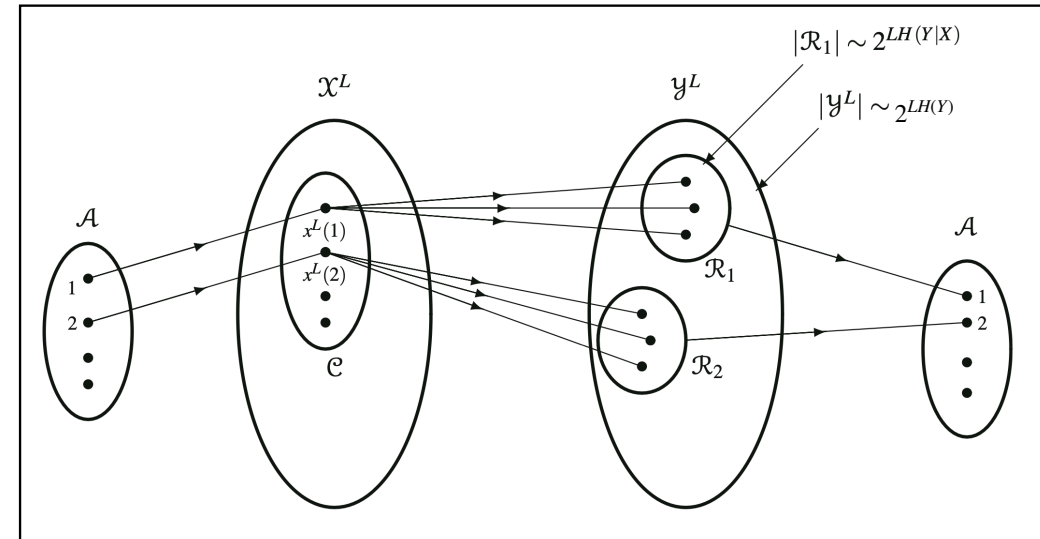


# Shannon's Channel Coding Theorem



- **Theorem** (Shannon's second theorem): Given a noisy channel with capacity  $C$  and a source with  $K$  messages, provided that  $\frac{1}{L} \log K < C$  there exists a coding and encoding system that guarantees an arbitrarily small error probability.
- Conversely, if  $\frac{1}{L} \log K > C$  no encoding scheme guarantees arbitrarily small error probability.
- **Proof:** Based on random codeword:  $x^L(a) = (x_1(a), x_2(a), \dots, x_L(a))$  with IID symbol and common distribution  $p_X(x)$
- Highly likely that  $Y^L$  lies in the typical set with size  $2^{LH(Y)}$
- If the input  $X^L$  is known then highly likely that  $Y^L$  lies in the conditional typical set of size  $2^{LH(Y|X)}$
- How many codewords can we have such that the outputs are distinguishable?
- Sphere packing problem and intuitively:  $\frac{2^{LH(Y)}}{2^{LH(Y|X)}} = 2^{LI(X;Y)}$
- The number of messages that can be faithfully transmitted is  $|M| = 2^{LI(X;Y)}$  with rate equal to  $\frac{\log_2 |M|}{L} = I(X; Y)$
- Optimize the rate by calculating  $R = \max_{p_X(x)} I(X; Y)$



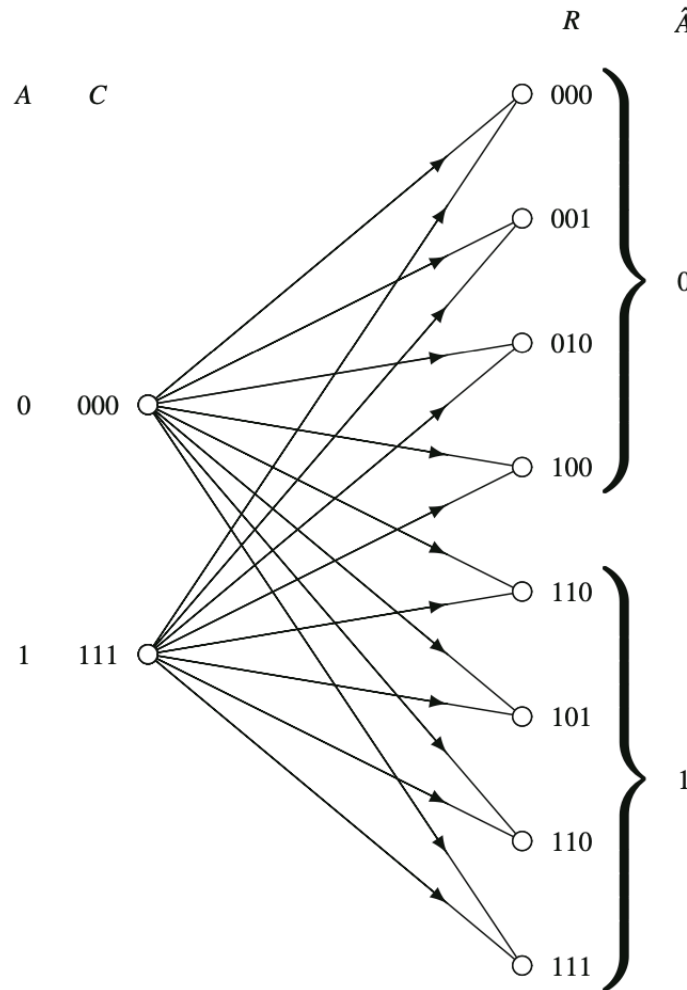


# Repetition Code Example

## • 3-1 repetition code

| $\mathcal{A}$ | $\mathcal{C}$ | $\mathcal{R}$ |
|---------------|---------------|---------------|
| 0             | 000           | 000           |
|               |               | 001           |
|               |               | 010           |
|               |               | 100           |
| 1             | 111           | 100           |
|               |               | 101           |
|               |               | 110           |
|               |               | 111           |

3-extension of of binary symmetric channel



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• Prob. of error:

$$P_{e3} = P[\hat{A} \neq A] = \binom{3}{2} \epsilon^2(1 - \epsilon) + \epsilon^3$$

• Let  $H(A) = 1$  bit/symbol (equiprobable symbols) and transmission rate:  $f_c = 1000$  symbols/s with  $\epsilon = 0.1$ .

• Without encoding: Information rate:  $H' = H(A)f_c = 1000$  bit/s and error prob.:  $P_e = \epsilon = 0.1$

• With 3-1 repetition code:

$$P_e := P[\hat{a} \neq a] = \binom{3}{2} \epsilon^2(1 - \epsilon) + \epsilon^3 = 0.028 \text{ with information rate: } H' \approx 333 \text{ bit/s } (f_s = f_c/3).$$

• Increasing repetitions, Intuitively,  $P_e \rightarrow 0$  provided that  $H' = f_s H \rightarrow 0$ .

• Shannon's counter intuitive conclusion:  $P_e \rightarrow 0$  provided that  $H' < C'$ .

• Here  $C' = f_c[1 + \epsilon \log \epsilon + (1 - \epsilon)\log(1 - \epsilon)] \approx 529$  bit/s

# Summary - I



- Information in a message is related to its randomness or uncertainty
- Information source is a random variable  $A$ , specified by a probability distribution  $p_A$
- Classical Shannon's Entropy (average information content):  $H(A) := - \sum_a p_A(a) \log_2 p_A(a)$  (in bits)
- In practice, information sources are *redundant*, while ideal sources have uniform probability distribution.
- **Data compression/source coding:** Transform redundant source to uniform source to minimise bits/message.
- Classical **Shannon's source coding theorem:** For a sequence of i.i.d r.v.  $(A_1, \dots, A_L)$ , for  $L \gg 1$ , the minimum number of bits to represent such a sequence is given by the entropy of the sequence.

# Summary - II



- **Channel coding** is used for reliable transmission of information over noisy channels.
- Classical coding: A source message of length  $L_0$  is encoded into  $L$ -long codeword  $W$  from a prefixed codebook.
- The noisy channel produces the stochastic mapping:  $W \rightarrow W'$  and Rate is defined as  $R = L_0/L$
- **Shannon's second channel coding theorem:** For an appropriate choice of a codebook, there is a threshold called channel capacity  $C$  s.t. any  $R < C$  is achievable with arbitrary small error probability.

**Thank You**

# References



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# CEPIITD

## Online PG Diploma: Advanced Communication Engineering with Quantum and AI Integration

Course: Quantum Information and Security

Instructor: Dr. Neel Kanth Kundu

Assistant Professor

Centre for Applied Research in Electronics (CARE)  
Bharti School of Telecom Tech. & Mgmt. (Joint Faculty)  
Indian Institute of Technology Delhi



# Quantum Information and Security

Module-2

## Quantum Information Theory

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# Overview



- Quantum Information Concepts
- Quantum Entropy
- Quantum Data Compression
- Quantum Channels
- Quantum Capacity



# Schmidt Decomposition

- Every pure state  $|\psi\rangle$  of a bipartite system (e.g., Alice + Bob) can be written in a special form called the **Schmidt decomposition**:

$$|\psi\rangle = \sum_{i=1}^d \lambda_i |i_A\rangle \otimes |i_B\rangle \text{ where:}$$

- $\lambda_i > 0$  are called **Schmidt coefficients**.
- $\{|i_A\rangle\}$  and  $\{|i_B\rangle\}$  are orthonormal bases.
- $\sum_i \lambda_i^2 = 1$ .
- $d \leq \min(\dim \mathcal{H}_A, \dim \mathcal{H}_B)$ , where  $\mathcal{H}_A$  and  $\mathcal{H}_B$  are the Hilbert spaces of the two subsystems.
- The number of non-zero  $\lambda_i$ 's is called the **Schmidt rank**.



# Schmidt Decomposition Example

- The Schmidt rank  $d$  tells us if a pure bipartite state is entangled.
  - If  $d = 1$ : the state is **not entangled** (product state).
  - If  $d > 1$ : the state is **entangled**.

- Example: Consider the Bell state  $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

- This can be written in Schmidt form:

$$|\psi\rangle = \lambda_1 |0_A\rangle |0_B\rangle + \lambda_2 |1_A\rangle |1_B\rangle \text{ with } \lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}.$$

- Since both coefficients are non-zero:  $d = 2 \implies$  State is entangled